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Metamaterials Break the Constraints of Traditional Civil Infrastructure

Amir H. Alavi

Advanced materials are essential for enhancing the resilience, efficiency, and adaptability of civil infrastructure. Yet, the core materials used in civil infrastructure systems today have remained largely unchanged for nearly a century, with fixed properties that limit their ability to realize these capabilities. Mechanical metamaterials offer a path to overcome the inherent constraints of conventional civil infrastructure materials. This perspective explores how mechanical metamaterials can redefine the backbone of civil infrastructure systems at the material level by encoding function in geometry, thereby enabling programmable mechanics and advanced multifunctional responses. Their scope of influence spans buildings, transportation networks, underground structures, offshore platforms, and architectural or interior systems, where thousands of components currently designed with conventional materials can be reimagined using optimized metamaterial counterparts. A mechanical metamaterial approach to civil infrastructure design can enhance structural performance by improving strength-to-density ratios, expanding the design space beyond conventional Ashby plot limits, and supporting applications such as seismic isolation, vibration damping, and lightweight construction. The discussion underscores the potential of mechanical metamaterials to introduce advanced functionalities such as energy harvesting, sensing, and wireless communication within civil infrastructure systems. These capabilities, traditionally associated with electronic devices, can be incorporated directly into structural components to create infrastructure with higher autonomy and adaptability. Multimodal integration with electromagnetic and photonic metamaterials further extends these possibilities by allowing materials to exhibit intrinsic communication, stealth behavior, or visual responsiveness. The perspective concludes with a roadmap outlining key developments and challenges toward a more integrated and responsive metamaterial-driven infrastructure. Within this framework, structurally tuned construction metamaterials can sense their environment, process information, and communicate with surrounding systems, functioning as the brain, eyes, and skin of future civil infrastructure.

1. Introduction

Civil infrastructure has long been shaped by conventional materials and fixed design standards. From bridges and tunnels to buildings and pavements, the civil infrastructure systems have largely relied on a familiar palette of bulk materials such as steel, concrete, and wood. These materials have supported the development of cities, transportation systems, and public utilities through their structural reliability and widespread availability.[1,2] However, continued reliance on these legacy materials and conventional design frameworks has resulted in systems that are increasingly misaligned with the demands of contemporary infrastructure.[3,4] Today's challenges, including urban densification, climate resilience, sustainability targets, and resource efficiency, are now coupled with emerging demands for infrastructure that supports connected technologies such as automated, connected, and electric vehicle systems.[5] These developments call for materials and structures that go beyond basic strength and stiffness, incorporating adaptability and compatibility with intelligent transportation and urban systems. One of the fundamental limitations in designing resilient and multifunctional civil infrastructure systems lies in the reliance on monolithic, compositionally defined materials. Once selected, these materials offer limited capacity for tuning performance or adapting behavior to changing conditions.[1,4,6–9] As a result, civil infrastructure components are often overdesigned to meet extreme load scenarios, leading to inefficient use of materials and excessive structural weight.[8] These components occupy valuable volume yet typically serve only one function, that is carrying load. In reality, they could be engineered to contribute significantly more. Structural elements may be configured to dissipate energy, harvest ambient forces, detect internal damage, or respond to environmental changes, all while continuing to fulfill their primary role in the structural system. The question is why this level of efficiency and multifunctionality has not yet made its way into mainstream civil infrastructure. The gap stems from material limitations and established design practices, further compounded by the challenges of uncertainty quantification

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and cost realism for new materials. Traditional materials are often selected for ease of use, availability, and standardization rather than adaptability or integration of new capabilities.[1] Design methodologies remain grounded in linear, 1D performance models.[10,11] In addition, fabrication of newer, functionally integrated materials often requires manufacturing techniques not yet common in civil construction, and their qualification depends on quantifying variability in properties, interfaces, and tolerances, as well as generating credible cost and maintenance data.[12] As a result, infrastructure systems have evolved with minimal change in how materials behave or how their roles are defined, despite rising expectations for more responsive and efficient solutions.

This disconnect between what civil infrastructure could achieve and what is currently delivered points to the need for a fundamentally new design approach. Mechanical metamaterials offer a path forward. Unlike traditional materials, the behavior of mechanical metamaterials is governed primarily by their engineered internal geometry rather than solely by their material composition.[13–18] This geometric control enables tailored mechanical and functional responses, allowing structural components to be programmed for specific tasks such as energy absorption, stiffness modulation, vibration damping, or real-time sensing.[14] These capabilities could be embedded without compromising the component's structural integrity. By shifting the focus from material composition to architectural design, mechanical metamaterials create opportunities to rethink the role of every physical element in the built environment. What has traditionally been viewed as passive and singular in function can now become active, efficient, and multifunctional. A wall no longer has to only carry load, it can also attenuate noise, harvest vibrational energy, or provide thermal buffering. A pavement system could adapt its compliance seasonally, sense fatigue damage in real time, or regulate drainage using embedded anisotropy. This transition has the potential to redefine how infrastructure is conceived, built, and operated, aligning material behavior with the complex and evolving needs of 21st-century civil and urban systems. This perspective presents a vision in which mechanical metamaterials overcome long-standing material and design constraints in civil infrastructure. It begins by outlining the fundamental shift from composition-based to geometry-driven design and highlights how this transition enables new forms of structural performance, including lightweighting and improved strength-to-density ratios. Applications are examined across key infrastructure domains, including buildings, transportation, underground, offshore, and architectural systems, with examples illustrating improved functionality and material efficiency. The discussion then expands to multifunctionality, focusing on embedded sensing, energy harvesting, and wireless communication, and considers how integration with electromagnetic (EM) and photonic metamaterials can further enhance capabilities. A roadmap of technical milestones and future challenges is then presented. The perspective concludes by building toward a vision of intelligent infrastructure, where construction materials function as responsive systems capable of actively interacting with their environment.

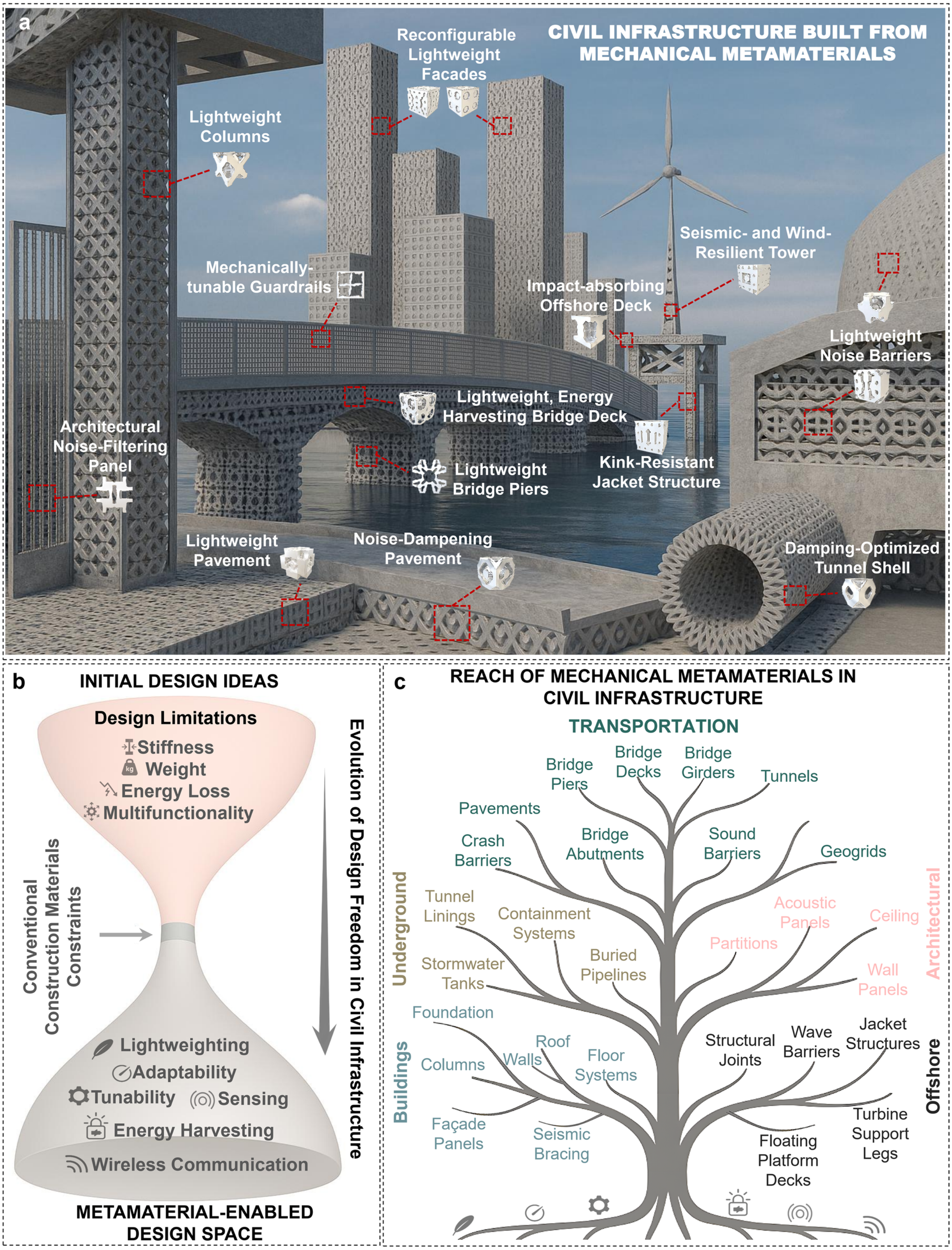
2. Scope of Influence of Mechanical Metamaterials in Civil Infrastructure

Mechanical metamaterials offer a fundamentally new approach to civil infrastructure design by shifting the focus from material composition to engineered internal geometry as the primary driver of mechanical behavior. **Figure 1a** illustrates a forward-looking vision for civil infrastructure reimagined through mechanical metamaterials. In this conceptual city, every major component, from buildings and pavements to tunnels, offshore platforms, and interior systems, is designed with engineered unit cell architectures that offer enhanced functionality, reduced weight, and improved adaptability. In contrast, Figure 1b illustrates how conventional construction materials impose a bottleneck on civil infrastructure design freedom. Their monolithic and compositionally fixed nature limits the ability to tailor performance parameters and hinders the realization of advanced functionalities. Mechanical metamaterials address these constraints by making topology a design variable.[13,14] This shift enables the integration of compliance, energy absorption, adaptability, sensing, and other advanced functionalities directly into structural components, moving from static material behavior to programmable mechanical response.[14] Such capabilities open new pathways for multifunctionality and can significantly expand the design space beyond what is possible with traditional construction materials. For civil infrastructure, which includes some of the world's largest and most enduring systems, this shift offers transformative potential. Yet despite these advantages, civil infrastructure remains one of the slowest sectors to adopt mechanical metamaterials. To explore this gap, I explore key infrastructure domains and demonstrate how mechanical metamaterials can enhance performance, reduce structural mass, enable embedded intelligence, and support entirely new functionalities.

Figure 1c provides a sector-based framework showing how mechanical metamaterials can serve as a unifying platform to enhance the performance of five core civil infrastructure domains: Buildings, Transportation, Underground, Offshore, and Architectural/Interior systems. As seen, the scope of influence is vast ranging from the small architectural elements to large-scale components such as bridge superstructures. Mechanical metamaterials offer a means to enhance the core functions of infrastructure elements by optimizing strength, reducing weight, and improving strength-to-density ratios through geometric design. This section outlines how metamaterials can elevate the inherent performance of each system, with an emphasis on lightweighting and structural efficiency.

2.1. Buildings Sector

In building structures, components such as façade panels, floor systems, foundations, load-bearing walls, seismic bracing, and structural columns can be redesigned using mechanical metamaterials. Unlike conventional materials chosen primarily for bulk properties like compressive strength or stiffness, mechanical metamaterials enable geometry-driven control over


a
CIVIL INFRASTRUCTURE BUILT FROM MECHANICAL METAMATERIALS
Reconfigurable Lightweight Facades
Lightweight Columns
Mechanically-tunable Guardrails
Impact-absorbing Offshore Deck
Seismic- and Wind-Resilient Tower
Lightweight Noise Barriers
Lightweight, Energy Harvesting Bridge Deck
Architectural Noise-Filtering Panel
Lightweight Bridge Piers
Kink-Resistant Jacket Structure
Lightweight Pavement
Noise-Dampening Pavement
Damping-Optimized Tunnel Shell
b
INITIAL DESIGN IDEAS
Design Limitations
Stiffness
Weight
Energy Loss
Multifunctionality
Conventional Construction Materials Constraints
Evolution of Design Freedom in Civil Infrastructure
Lightweighting
Adaptability
Tunability
Sensing
Energy Harvesting
Wireless Communication
METAMATERIAL-ENABLED DESIGN SPACE
c
REACH OF MECHANICAL METAMATERIALS IN CIVIL INFRASTRUCTURE
TRANSPORTATION
Bridge Piers
Bridge Decks
Bridge Girders
Tunnels
Pavements
Crash Barriers
Bridge Abutments
Sound Barriers
Geogrids
Underground
Tunnel Linings
Containment Systems
Stormwater Tanks
Buried Pipelines
Architectural
Acoustic Panels
Ceiling
Partitions
Wall Panels
Buildings
Foundation
Columns
Walls
Roof
Floor Systems
Façade Panels
Seismic Bracing
Offshore
Structural Joints
Wave Barriers
Jacket Structures
Turbine Support Legs
Floating Platform Decks

mechanical behavior of these components. This allows the building components to simultaneously support load, dissipate energy, and adapt to vibration or environmental forces. For instance, walls and slabs can be constructed from architected concrete, metal or polymer lattices with tailored internal geometries that optimize stress distribution and reduce weight. Functionally graded metamaterials enable localized stiffness control by combining rigid and compliant regions within a single element.[19] This approach is especially effective at corners, interfaces, and high-stress areas of buildings where conventional reinforcement is difficult to apply. Floor systems could be designed with lightweight lattice topologies, such as the octet lattices.[20,21] This strategy can result in achieving lower dead loads without compromising flexural rigidity, making them ideal for multistory applications. Similarly, façade panels with embedded lattice reinforcements can be used to minimize material use while maintaining performance under wind and gravity loads.

2.2. Transportation Sector

Transportation infrastructure, including bridges, pavements, geogrids, tunnels, and noise barriers, is constantly subjected to mechanical stress, fatigue, environmental wear, and dynamic loads. Bridges, especially long-span or cable-stayed types, are limited by dead load and dynamic response under wind or traffic-induced vibrations.[22,23] Integrating lightweight lattice metamaterials into bridge decks or piers can significantly reduce self-weight while maintaining stiffness and strength. For example, 3D-printed concrete or polymeric lattices with tailored topology can replace internal fillers or nonstructural layers (e.g., ref. [24,25]). Mechanical metamaterials can be incorporated into the surface, base, or subbase layers of pavement systems. By replacing bulk material with load-directed cellular geometries (e.g., refs. [7,8]), this approach can increase strength-to-density ratios, allowing pavements to support heavy loads while reducing weight and material use. Substructures such as base and subbase layers can also be redesigned using metamaterial patterns to lower thickness. With growing interest in precast methods in pavement engineering,[26] prefabricated metamaterial-based panels offer a promising solution for accelerated construction in both urban and remote settings. For example, a metamaterial-inspired base layer with a hexagonal lattice may reduce aggregate volume while maintaining stability under truck axle loads, potentially allowing faster compaction and easier transport to remote sites. Mechanical metamaterials can be deployed to improve the strength-to-density ratio in tunnel linings via architected geometries that distribute loads efficiently with reduced mass. This enables thinner, structurally sound elements that lower overall material usage. The use of lightweight, prefabricated panels or rings simplifies transport, handling, and installation in tight underground spaces.[27] Reduced component weight also decreases excavation volume and accelerates construction timelines. For instance, a tunnel lining system made of interlocking panels with embedded 3D lattice cores could significantly reduce the overall ring weight (e.g., Figure 1a). This approach can be especially suitable for micro-tunneling and soft-soil conditions, where ease of handling and settlement control are essential. Geogrids are widely used in pavement systems to reinforce base and subbase layers, improve load distribution, and reduce deformation. By applying mechanical metamaterial principles, geogrids can be engineered with architected geometries that enable high tensile performance with reduced material usage. They can also be made from recycled plastic waste, with the metamaterial design compensating for material variability or lower mechanical quality by tailoring the geometry to meet performance needs. For example, a geogrid with a re-entrant honeycomb pattern made from recycled polyethylene may offer similar pullout resistance using less material than conventional punched-sheet designs. Its rollable form also supports faster on-site deployment, contributing to more efficient and sustainable pavement systems. Noise barriers are another critical components in transportation infrastructure for mitigating traffic noise along highways and rail corridors. These barriers can be prefabricated using lightweight architected geometries that maintain structural integrity while significantly reducing weight. The reduced mass simplifies transport and installation, making them ideal for large-scale deployment along elevated or difficult-to-access corridors. For instance, in a recent study, we demonstrated that cementitious mechanical metamaterial acoustic blocks can serve as effective noise barriers, exhibiting a remarkably high strength-to-density ratio of 0.086 MPa m^3 kg^{-1}, far exceeding that of traditional cementitious materials.[7,8] Notably, these blocks achieved a sound transmission coefficient only 2% higher than that of solid concrete control specimens.

2.3. Underground Sector

The underground sector includes essential components such as buried pipelines, stormwater tanks, protective enclosures, utility

Figure 1. Vision for mechanical metamaterial-driven civil infrastructure. a) A futuristic vision where civil infrastructure is entirely constructed from mechanical metamaterials. This concept reimagines buildings, transportation networks, underground systems, offshore structures, and architectural elements guided by metamaterial design principles. Each component leverages tailored unit cell geometries to enable advanced mechanical performance, such as high strength-to-density ratios, dynamic response control, and integrated multifunctionality. Columns and piers are optimized for strength with minimal mass; pavements integrate noise-dampening lattices and energy harvesting capability; guardrails are mechanically tunable; tunnel shells are designed for damping and soil pressure control; jacket structures resist kinking under stress; offshore decks absorb impact; towers are resilient to seismic and wind loads; facades are lightweight and reconfigurable, and lightweight noise barriers and architectural panels filter unwanted sound. Note: The infrastructure systems shown in this figure were created through periodic assembly of unit cells positioned nearby, each explored using the author's in-house generative metamaterial design algorithm. These designs are schematic representations meant to illustrate core concepts, not precise engineering models. b) Evolution of design freedom in civil infrastructure. Conventional construction materials impose strict limitations on civil infrastructure design due to their compositionally fixed and monolithic nature, forming a bottleneck that constraints performance tuning and multifunctionality. Mechanical metamaterials, by contrast, open a significantly expanded design space. c) Sector-based taxonomy illustrating the breadth of civil infrastructure domains that can potentially be impacted by mechanical metamaterials. The framework spans five core sectors: buildings, transportation, underground, offshore, and architectural/interior systems. Each contains subdomains where metamaterial-enabled capabilities such as lightweighting, damping, adaptability, and other advanced functionalities may be deployed.

conduits, tunnel linings, and containment systems. These elements are often subjected to high earth pressure, dynamic loads, and limited access during installation, making weight, strength, and durability critical design factors.[28,29] Prefabricated metamaterial components, such as interlocking lattice panels, can simplify transport and assembly in tight or deep excavation sites. As in the transportation sector, tunnel linings can incorporate internal architected voids to reduce material use without sacrificing strength. Likewise, stormwater tanks made from metamaterial-enhanced polymer composites can be significantly lighter than conventional designs while still supporting equivalent overburden loads.

2.4. Offshore Sector

The offshore sector includes key structural systems such as wave barriers, floating platform decks, jacket structures, turbine support legs, and structural joints, all of which face demanding mechanical and environmental loads. These elements are typically constructed with heavy, overbuilt components to ensure stability against waves, wind, and corrosion.[30] Mechanical metamaterials enable a significant improvement in strength-to-density ratio of these elements by embedding architected geometries that redirect stresses while reducing overall mass. This lightweighting translates to lower material usage, easier transport to offshore sites, and reduced buoyancy or foundation demands. For instance, a turbine support leg incorporating a periodic truss lattice may reduce steel volume usage while preserving bending stiffness, reducing lift vessel requirements during installation. Similarly, wave barriers formed from modular, prefabricated metamaterial panels with energy-absorbing cores can maintain impact resistance while reducing weight for quicker deployment.

2.5. Architectural and Interior Sector

This sector, though sometimes viewed as distinct from core structural systems, includes functional components that are integral to civil infrastructure, particularly in large-scale buildings, transit hubs, and public facilities.[31–33] Elements such as acoustic panels, floor systems, wall panels, ceiling elements, furniture, handrails, partitions, and decorative components contribute to occupant comfort, spatial organization, and performance under usage demands. These components can be optimized using mechanical metamaterials to be both lightweight and structurally efficient. Lightweighting is especially valuable in multistory buildings or retrofit applications where reducing nonstructural mass can lower seismic demand and ease installation.[34,35] For example, ceiling panels with lattice-based metamaterial cores can weigh significantly less than conventional gypsum tiles while maintaining impact resistance and acoustic dampening. Wall partitions made from folded or cellular polymer structures can combine stiffness and flexibility, improving safety and mobility in public settings. Prefabricated metamaterial-based interior components not only reduce load but also simplify transport and assembly in complex architectural spaces. These advantages make architectural and interior elements a meaningful part of the civil infrastructure ecosystem, especially when designed to meet both aesthetic and mechanical performance criteria.

3. Ashby Plot Shift Enabled by Mechanical Metamaterials

All the strength-to-density gains discussed across sectors, whether in pavements, tunnels, offshore platforms, or architectural systems, reflect more than performance improvements. They represent a fundamental shift in how construction materials behave and how their capabilities are designed. Traditionally, material selection guided by Ashby plots[36] has involved picking from a fixed set of options based on specific stiffness or strength-to-density ratio, often requiring tradeoffs between weight, strength, and cost. **Figure** 2a presents an Ashby plot illustrating the relationship between compressive strength and density for a range of construction materials. The gray region represents the typical performance envelope of conventional civil infrastructure materials. Concrete, from standard Portland mixes to high-performance (HPC) and ultra-high-performance (UHPC) variants, exhibits compressive strengths of 20 MPa to over 150 MPa and densities of 1600–2500 kg m^{-3}, corresponding to strength-to-density ratios of 0.008–0.060 MPa m^3 kg^{-1}.[37–39] Metals such as structural steel (e.g., ASTM A36, A992) offer tensile strengths between 400 and 550 MPa at a density of ≈7850 kg m^{-3}, yielding ratios around 0.051–0.070 MPa m^3 kg^{-1}. Aluminum alloys (e.g., 6061-T6), with strengths of 250–300 MPa and densities near 2700 kg m^{-3}, exhibit higher ratios of 0.093–0.111 MPa m^3 kg^{-1}. Wood, particularly in engineered forms like cross-laminated timber, remains vital for mid-rise and residential construction.[40,41] Engineered timber products, such as cross-laminated timber, typically achieve ratios of 0.080–0.160 MPa m^3 kg^{-1}. Polymers and polymer composites, including HDPE, PVC, and FRPs, span tensile strengths of 30–500 MPa and densities of 900–2000 kg m^{-3}, yielding a broad range of ratios from 0.015 to 0.333 MPa m^3 kg^{-1}. Traditional ceramics used in masonry, such as clay bricks and terracotta, have compressive strengths of 10–80 MPa and densities of 1600–2200 kg m^{-3}, with corresponding ratios of 0.006–0.036 MPa m^3 kg^{-1}

Despite this diversity, all of these material classes remain compositionally fixed. Thus, their performance is ultimately limited by molecular structure and atomic bonding. As shown in Figure 2a, mechanical metamaterials enable a fundamental shift in the Ashby material property space, allowing construction materials to move leftward on the strength–density plot (green region). This shift on the Ashby chart is not merely symbolic and reflects a key advantage of mechanical metamaterials. It suggests that civil infrastructure components made from metamaterials can deliver comparable or adequate mechanical performance at significantly lower densities, thereby enhancing specific strength. For instance, a concrete panel that once weighed 800 kg can now weigh 500 kg without significantly losing performance. A bridge component can span longer with less steel. An interior panel can absorb sound, respond to vibrations, and carry load, all at once. These transformations are supported by experimental evidence and are no longer hypothetical. For example, our recent work[7,8] demonstrated that a mechanical metamaterial design for concrete can reduce density by 20%–40% while maintaining compressive strength within 10% of the original material. The Ashby plot shift is evidently not confined to any one construction material. Whether in concrete, steel, wood, or polymers, mechanical metamaterials allow for

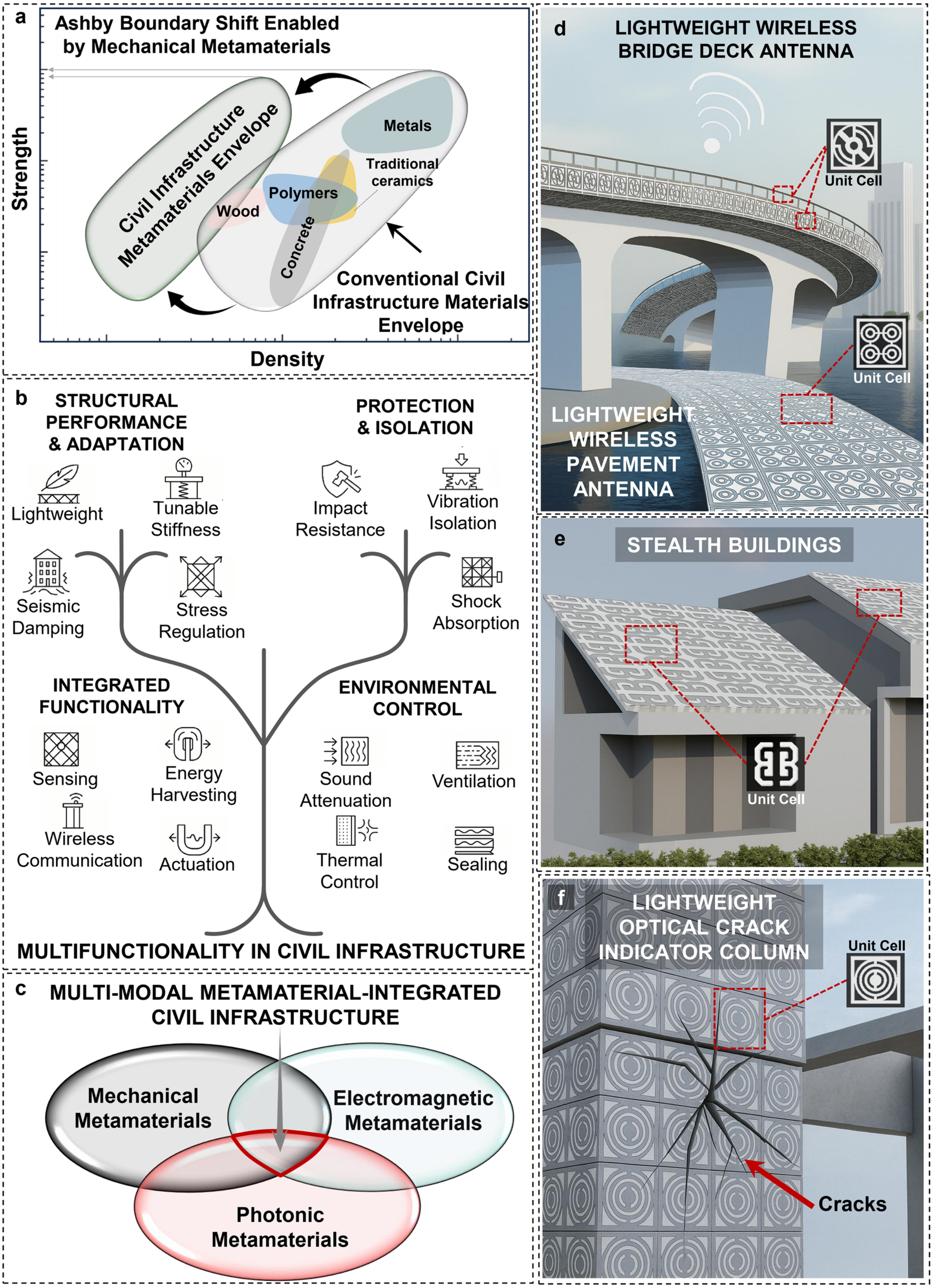


Figure 2. Advanced civil infrastructure capabilities enabled by mechanical, EM, and photonic metamaterials. a) Ashby-style material property plot comparing strength and density of conventional civil infrastructure materials (concrete, metals, wood, polymers, and traditional ceramics) with mechanical metamaterial counterparts. By leveraging architected geometry rather than composition alone, mechanical metamaterials shift the boundary of achievable construction material performance, enabling combinations of low density and high strength that exceed traditional bulk material limits. This shift

performance enhancement without material substitution. Hybrid systems extend this potential even further. For instance, hybrid mechanical metamaterial-concrete-steel composites could combine lightweighting with energy absorption and multifunctionality. These configurations allow civil components to move into regions of the Ashby chart once reserved for aerospace or high-end composites. Such metamaterial-driven civil infrastructure systems can further evolve from fixed-function systems to programmable structures.

4. Multifunctionality beyond Specific Strength

Civil infrastructure has traditionally been designed with a primary focus on strength. However, as modern infrastructure is increasingly expected to adapt to environmental demands, support data-driven systems, and integrate sustainability, the role of structural materials must evolve. Mechanical metamaterials are uniquely suited to address this shift. These architected materials can deliver more than mechanical performance. They can sense, respond, harvest energy, regulate temperature, attenuate sound, and even communicate with external systems.[4,14] This kind of multifunctionality, made possible by geometry, represents a clear shift from traditional ways of thinking about materials. Figure 2b presents four key benefit categories and their associated features, with conceptual examples provided to illustrate each, as follows:

4.1. Structural Performance and Adaptation

This category includes features such as lightweighting, tunability, and adaptive stiffness. It is broadly applicable, especially in buildings, transportation, and architectural/interior sectors, where reducing material weight and enhancing mechanical response can improve load-bearing efficiency and adaptability. Mechanical metamaterials enable novel forms of structural adaptation through their internal geometries.[14] For example, a slab embedded with auxetic cores (e.g., ref. [42]) or negative stiffness inclusions (e.g., ref. [43]) can exhibit superior damping properties, reducing floor vibrations in tall or long-span buildings. In multistory construction, this translates to thinner slabs and reduced foundation loads without compromising performance. Seismic resilience is another area of strong impact. Embedding auxetic layers into shear walls or braces can allow energy to be passively dissipated during earthquakes, reducing peak accelerations transmitted to the rest of the structure. With growing interest in adaptive building facades,[44] these elements can be broadly designed as metamaterials. Examples include origami-inspired configurations used as responsive louvers for daylight control (e.g., ref. [45]), or panels with anisotropic stiffness that adapt under wind loads, limiting wind-induced vibration and serviceability issues. Façade elements that integrate structural framing with acoustic metamaterial cores can function as multifunctional systems, supporting wind and gravity loads while also filtering sound. Another example is metamaterial panels in high-rise buildings that incorporate embedded compliant mechanisms, allowing them to deform under lateral forces and reduce peak wind pressures. In transportation, mechanical metamaterial-based pavement sub-base layers can be tailored to adapt to moisture and temperature gradients, helping to reduce freeze-thaw cracking. In tunnels or bridge decks, using hollow-core, lightweight metamaterial modules can result in flexible deformation to accommodate dynamic loads or temperature shifts. The tunnel linings could also be structured as variable-stiffness lattices to accommodate ground movement. Floating offshore decks can benefit from lightweight metamaterial sandwich panels (e.g., 3D kagome or re-entrant honeycomb panels) optimized for out-of-plane stiffness resistance. The ability to fine-tune stiffness locally enables structural modules to respond to varying load profiles more effectively than conventional monolithic materials. However, adaptive features should not be credited toward life safety. The primary structure should satisfy strength and drift limits with adaptation disabled. Components should be designed to default to a safe, high-stiffness state when stroke, strain, temperature, or pressure thresholds are exceeded, incorporating force-limiting fuses or slip mechanisms where appropriate. Interfaces should be protected using compliant sleeves, shear keys, and surface profiling to prevent debonding, and details should ensure ductile, controlled degradation. Verification should be performed at the subassembly scale under three conditions: system enabled, system disabled, and system deliberately damaged. Only configurations that meet ultimate and service limits in the disabled state, and that avoid brittle failure or loss of load path in the damaged state, should be advanced to pilot implementation.

4.2. Protection and Isolation

This category covers features such as shock absorption, impact resistance, and vibration isolation. It is especially relevant to Underground and Offshore systems, and to Architectural/Interior applications requiring vibration damping. In underground settings, using auxetic geometries in buried pipelines or nested lattice shells in stormwater tanks can help distribute stress and prevent crack propagation. Folded origami metamaterials can

supports new design paradigms for lightweight, high-performance infrastructure systems. b) Multifunctionality tree of mechanical metamaterials in civil infrastructure organized into four key domains: structural performance and adaptation, protection and isolation, integrated functionality, and environmental control. Each category highlights core capabilities that can be embedded into infrastructure through geometry-driven design. This multifunctional integration redefines how infrastructure can sense, adapt, protect, and interact with its environment. c) Venn diagram showing interaction of mechanical, EM, and photonic metamaterials in a unified civil infrastructure system. Conceptual illustrations highlighting the transformative potential of multimodal metamaterial integration in civil infrastructure: d) bridge deck and pavement systems engineered as wireless structural antennas, either by embedding periodic conductive or dielectric inclusions into the concrete matrix or by applying EM stencil-patterned coatings on the surface, e) a stealth-enabled roof structure designed with EM cloaking metamaterials to suppress signal visibility, enhancing privacy or defense, and f) artificial intelligence (AI) load-bearing column with integrated photonic metamaterials that visually indicate internal strain or cracking through optical responses, enabling passive structural health monitoring. Note: The infrastructure systems shown are schematic representations assembled using randomly created EM-like unit cells. These designs are intended to illustrate conceptual functionality and do not represent actual engineered or simulated models.

be used in shelters and protective enclosures to absorb blast loads, collapsing in a controlled manner without transmitting shock to the interior. This principle has promising applications in civil shelters and defensive infrastructure.[46,47] Offshore platforms could be designed with auxetic inserts into pile sleeves and jacket nodes to reduce stress concentrations from wave and current oscillations. Underground infrastructure, including pipes and tanks, can benefit from mechanical metamaterial designs that resist ovalization and joint separation under soil loads or seismic vibration. For instance, tank shells with hierarchical lattice patterns can absorb both internal and external pressure more efficiently while reducing crack propagation. In interior environments such as healthcare settings, vibration isolation is critical to preserve the precision and functionality of sensitive equipment.[48] In this context, mechanical metamaterial floor systems incorporating embedded negative stiffness elements can provide targeted isolation across specific frequency ranges. Floors using bandgap engineered lattices can block vibrations in a direction-dependent manner, enabling one-way vibration transmission to reduce noise in multistory structures.

4.3. Integrated Functionality

This category includes sensing, wireless communication, energy harvesting, and actuation functionalities. These features are increasingly critical in Transportation, Offshore, and Architectural/Interior domains, where infrastructure must interact with its environment, report damage, or power embedded electronics without external energy sources.[49–51] Mechanical metamaterials can be inherently tailored to respond to deformation, making them well suited for use as embedded sensors.[4,52,53] Unlike traditional structural sensors that are surface-mounted and often require wiring, mechanical metamaterial-based sensors can be volumetric, integrated within the structural body, and operate without electronics. For instance, origami-inspired triboelectric nanogenerators generate electrical signals when mechanically deformed,[54,55] enabling volumetric, wire-free sensing of strain, compression, or vibration. Such functional elements can be seamlessly integrated into pavements as geogrids, bridge decks, or structural frames to detect damage or usage patterns. Piezoelectric skins patterned with auxetic or hexagonal motifs (e.g., refs. [56,57]) may also resolve load profiles spatially in real time. Mechanical metamaterials can also function as energy harvesters, converting waste mechanical energy into electrical power (e.g., refs. [52,53,58,59]). Dynamic forces from vehicular movement, pedestrian traffic, wind, or water flow are traditionally dissipated. Metamaterial geometries can localize strain into specific regions where nanogenerators are embedded. In pavements, compliant metamaterial sublayers with embedded triboelectric nanogenerators or piezoelectric elements can power streetlights or embedded sensors. Bridges and façades exposed to wind can use fluttering metamaterial fins with integrated electromechanical interfaces to capture aeroelastic energy. Water infrastructure can benefit from metamaterial surfaces with resonant cavities that scavenge energy from turbulence or vibration. Mechanical metamaterial energy harvesters can operate in various modes, such as bending, shear, compression, or twist, increasing their viability across civil infrastructure domains.

Perhaps the most transformative potential lies in enabling "talking civil infrastructures." Mechanical metamaterials, while traditionally associated with structural performance, can be inherently designed to exhibit wireless functionality. As demonstrated in our recent work on meta-mechanotronic systems,[60] mechanical metamaterials can serve as self-powered, electronic-free wireless communication platforms. When subjected to strain or vibration, these systems generate a strain-induced polarization electric field, enabling signal transmission without external power.[60] We demonstrated this concept using composite metamaterial implants that acted as integrated sensing units, harvesting energy, detecting force, and wirelessly transmitting data up to ≈20 cm, limited mainly by charge accumulation and discharge rate.[60] Other studies have achieved longer transmission ranges using higher-power triboelectric nanogenerators, transmitting over 30 m in air,[61] and enabling text or image transmission at 16 bit s^{-1} over 100 m through a water conduit.[62] Although these demonstrations were conducted in biomedical, wearable, and underwater contexts, the same principle can be extended to civil infrastructure. Such materials could serve as the "beating hearts" of future structures, communicating their state through applied mechanical forces without external electronics. With appropriate scaling and integration, practical transmission distances in civil systems could range from sub-meter levels through reinforced media to several meters in open or conductive environments. Embedded or surface-mounted conductive layers, reinforcing steel, or metallic joints can act as passive receivers to enhance coupling efficiency. At this stage, this type of wireless communication should be viewed as a complement to conventional sensor networks rather than a replacement, providing self-powered event flags, wake-up triggers, or low-bit identifiers that reduce wiring and maintenance requirements. Future studies should focus on optimizing coupling geometry, minimizing interference from grounding and rebar networks, and validating reliability through field-scale range and bit error rate testing.

4.4. Environmental Control

This category includes features such as sound attenuation, thermal control, ventilation, and sealing. These functions are critical in Buildings, Transportation, and Architectural/Interior domains, where acoustic performance, air regulation, and environmental buffering are essential. Phononic metamaterials can be considered a class of mechanical metamaterials tailored to control mechanical waves.[63–65] In this context, they are treated as an umbrella category that includes both phononic crystals and acoustic metamaterials, regardless of the specific dispersion mechanism involved. By embedding resonant cavities, mass-in-mass structures, or labyrinthine channels into walls or ceilings, these systems can achieve remarkable attenuation of sound. Hybrid panels combining such metamaterials with structural load-bearing cores can reduce material layering and improve integration. These are especially effective in sound barriers, tunnel linings, and bridge parapets where noise reduction is needed without sacrificing strength. In interior applications, Helmholtz resonator-based metamaterials (e.g., ref. [66]) can allow precise control of acoustic environments in offices, homes, or public spaces.

Thermal control and ventilation are closely linked in mechanical metamaterial systems.[67,68] Mechanical metamaterials with tailored porosity or gradient architecture can either resist or facilitate heat flow depending on external conditions, while also modulating airflow pathways (e.g., ref. [69]). For example, gradient metamaterial slabs may be engineered to direct heat flow directionally, reducing HVAC loads, while porous channels can enable passive ventilation through building façades or structural skins. Phase-change materials can be embedded within metamaterial voids to buffer diurnal thermal swings in walls or flooring. At the same time, origami-based metamaterial panels can passively regulate airflow by opening or closing in response to temperature-induced expansion, combining both ventilation and thermal responsiveness. These dual-mode functions reduce the need for mechanical systems and improve overall energy efficiency. In underground systems, capillary-tuned metamaterial channels can redirect moisture and airflow, buffer freeze-thaw cycles, and help ventilate otherwise sealed environments, protecting linings and improving service life. Mechanical metamaterial principles hypothetically enable structural panels that integrate thermal regulation, ventilation, sound control, and load-bearing capacity in one system. For example, in interior applications, metamaterial walls or partitions could be designed to dynamically modulate both airflow and acoustic properties based on occupancy. Panels with anisotropic stiffness can deform to allow adjustability in doors, movable partitions, or even tactile feedback in handrails. Visually exposed lattice structures serve both decorative and functional purposes, enabling light diffusion, sound attenuation, ventilation, and thermal buffering in a single integrated design. Also, mechanical metamaterials can potentially enhance sealing performance. For example, pressure-responsive or interlocking metamaterial geometries can be incorporated to expand, collapse, or self-tighten under load, maintaining an effective seal under varying pressure or temperature conditions. This is especially useful in joints, gaskets, underground linings, or panel interfaces where traditional seals may degrade or fail.

5. Toward a Unified Metamaterial Framework for Civil Infrastructure

While mechanical metamaterials can lead the way in redefining how civil infrastructure systems manage loads, vibrations, and multifunctionality, they are only one component within the broader landscape of architected material systems. Other major classes of metamaterials, including EM and photonic metamaterials, offer equally transformative potential, especially when integrated with mechanical systems in hybrid, multifunctional designs. Combining these metamaterial types (Figure 2c), mechanical, EM, and photonic, can enable multimodal civil infrastructure that can sense, adapt, protect, and interact. When a pavement functions as a sensor, a signal director, and a structure, the process shifts from constructing architecture to embedding “intelligence” into matter at the micro and mesoscale. The following section highlights some of these possibilities, ranging from near-term applications to more speculative example involving EM and photonic metamaterials in civil infrastructure systems.

5.1. Integration of EM Metamaterials into Infrastructure

EM metamaterials are engineered to manipulate EM waves in non-conventional ways, including negative refraction, cloaking, bandgap filtering, and signal focusing.[70,71] EM wave control can become essential for next-generation civil infrastructure, particularly in smart cities and automated transportation networks. Buildings and tunnels commonly interfere with wireless signals, leading to dead zones or degraded connectivity.[72,73] This is an increasingly problematic issue as civil infrastructure becomes more digitized and reliant on real-time data exchange. Building façades, walls, and tunnel linings can be engineered directly as EM metasurfaces to direct or amplify 5G and WiFi signals. This could eliminate the need for active relay systems. These metasurfaces can redirect signals using passive, geometry-controlled reflection and transmission patterns. They can even be programmed to change their EM response dynamically to focus signals toward specific locations or users. A conference room wall, for instance, could direct signals to a table of devices during a meeting, then suppress them to enhance privacy. This can be achieved entirely through structural tuning without any embedded electronics. In control rooms, hospitals, or data centers embedded within buildings, EM interference is a growing concern.[74,75] Walls in these buildings can serve as load-bearing EM metamaterials, supporting structural loads while selectively blocking or absorbing specific frequency bands to filter out unwanted signals and allow others to pass. These materials could be applied as architected coatings or embedded interlayers within walls and floors, creating frequency-selective shielding zones, without the weight and expense of traditional Faraday cages.[76] Multifrequency metamaterial panels can function as smart filters, suppressing nearby drone signals, preserving GPS accuracy during outages, and isolating sensitive communication networks during critical operations.

One of the most compelling directions is the fusion of structural materials with antenna functionality. For example, a bridge deck or concrete pavement could be engineered as a large-scale metamaterial antenna, with its concrete or composite structure engineered to relay traffic sensor data, environmental conditions, or communication signals. As shown schematically in Figure 2d, the antenna function is not embedded within a separate component. The bridge or pavement structure itself performs as the antenna. This could hypothetically be realized by embedding periodic conductive or dielectric inclusions into the concrete matrix, creating a structured material that interacts with EM waves and functions as a large-scale passive antenna. Alternatively, the EM functionality can be introduced through stencil-patterned coatings applied directly to the surface. Using masking techniques, these coatings composed of unit cell arrays can be deployed over large areas without modifying the internal structure. A relevant design strategy was demonstrated by Singh et al.,[77] who developed a low-cost additive process for fabricating metamaterial microwave absorbers using a patterned stencil mask and conductive paint. While their work was not focused on civil infrastructure, the same approach could arguably be adopted using appropriate coating materials. Spray-based or roll-on deposition guided by reusable stencil masks would make this method practical for scalable field deployment. An important consideration is to ensure structural integrity and long-term durability when integrating conductive inclusions or patterned coatings into load-bearing elements. One viable approach is to treat these features as nonstructural components and position them in low-strain regions to minimize stress concentration. Interfaces may include compliant interlayers or surface profiling to maintain bond performance and prevent delamination under thermal or moisture cycling. Surface patterns can be recessed beneath protective overlays such as concrete or polymer coatings to reduce point-load and wear effects. Electrical isolation between dissimilar materials and proper sealing of penetrations can help mitigate galvanic and moisture-driven degradation. For initial implementations, surface-applied and replaceable patterns are preferable, enabling inspection and maintenance using nondestructive evaluation methods.

In military or sensitive urban installations, signature control is critical. Using metamaterials with cloaking or scattering-suppression properties, it is possible to design walls or shelters that appear electromagnetically neutral, effectively concealing themselves from radar or other detection systems. These electromagnetic stealth structures may be configured using layered metasurfaces and applied to embassy bunkers, critical data vaults, or drone-protected buildings. This vision is schematically shown in Figure 2e. Though still speculative, such stealth structures could be designed to adjust their EM response dynamically by reconfiguring unit cell geometries or using tunable materials, making them visible to authorized scans while remaining hidden from others.

5.2. Integration of Photonic Metamaterials into Infrastructure

Photonic metamaterials operate in the visible to near-infrared spectrum, allowing control over light in ways that are not possible with traditional glass or coatings.[78,79] Their relevance to civil systems lies in their ability to enable thermal control, visibility management, passive cooling, and adaptive optics. As cities face rising temperatures and the growing impact of the urban heat island effect,[80] buildings play a major role in excess heat retention. Photonic metamaterials could be tailored to reflect sunlight while simultaneously releasing heat through radiative cooling. Roofing, pavements, and wall panel façades could serve as photonic load-bearing structural elements to perform these functions and help reduce cooling demands.

Structural health monitoring through optical behavior presents another major opportunity. By integrating optical metasurfaces, photonic crystals, or tunable reflectivity elements into construction materials, it becomes feasible to create visual indicators that respond to mechanical strain or cracking. These embedded optical features can modulate reflected or transmitted light in response to deformation, enabling passive, real-time visualization of structural stress or damage without external power or wiring. For instance, an optical metamaterial wall could change color when overstressed, while a lightweight metamaterial bridge deck might display a visible interference pattern as microcracks begin to form (Figure 2f). Another example is high-rise metamaterial façades that could visually indicate stress buildup before failure. Unlike conventional sensors, these indicators require no power or data transmission, functioning instead as passive material warning signals.

In summary, combining mechanical, EM, and photonic metamaterials opens the door to infrastructure capabilities that once seemed like science fiction. The physics and design principles are largely in place. The next step is to integrate them through cross-disciplinary collaboration and move these materials from the lab to real-world use.

6. Roadmap and Key Challenges Ahead

The next 10–20 years hold strong potential to reshape civil infrastructure through the integration of mechanical metamaterials. In this emerging paradigm, construction materials are no longer passive or static. Instead, they are engineered to sense, respond, and adapt to their environment over time. **Figure 3a** illustrates a Gartner-style Hype Cycle for mechanical metamaterials, emphasizing their projected 20-year trajectory in civil infrastructure. The mechanical metamaterials field began in the late 1980s with Roderic Lakes' auxetic foam exhibiting negative Poisson's ratio.[81] The 1990s saw theoretical expansion around auxetics and nontraditional mechanical behavior.[82–86] In the 2000s, interest shifted to lattice architectures and topology-optimized geometries for stiffness and strength tuning.[87–92] During the 2010s, advancements in additive manufacturing enabled the fabrication of complex micro-architectures, sparking interest in applications such as programmable stiffness, shape morphing, and impact mitigation.[93–111] Yet this momentum slowed as the field encountered two critical obstacles: fabrication limitations and scalability concerns. In the 2020s, the development of multifunctional mechanical metamaterials gained momentum, with a focus on integrating sensing, energy harvesting, and computing to enable intelligent behavior.[55,60,111–125] Around the same time, data-driven and artificial intelligence (AI)-based inverse design emerged to accelerate the discovery of high-performance metamaterial architectures.[126–138] The field is entering a more deliberate, systems-level phase enabled by AI tools and scalable digital fabrication platforms, as illustrated in Figure 3a. However, it was not until 2014 that the potential use of mechanical metamaterials in civil engineering began to emerge, with a few scattered studies exploring how resonant or periodic elements could serve as seismic metamaterials to shield buildings from surface waves when embedded in soil.[139–144] These early efforts were followed by limited investigations into cementitious lattices for lightweight structures (e.g., refs. [24,145–151]), initial explorations of architected microstructures for load-bearing structural applications (e.g., refs. [152–159]). Yet, the combined use of the terms "mechanical metamaterial" and "civil infrastructure" did not appear in the published scientific literature until 2023,[4] highlighting how recently this area has emerged and how much untapped potential remains within the field. To reach full deployment within civil infrastructure, three key challenges remain: (1) efficient design space navigation, (2) scalable and cost-effective fabrication, and (3) seamless integration with traditional infrastructure systems. The following sections outline each of these in detail.

6.1. Navigating an Expansive Search Space

Unlike conventional materials, the properties of mechanical metamaterials come from their geometry rather than merely

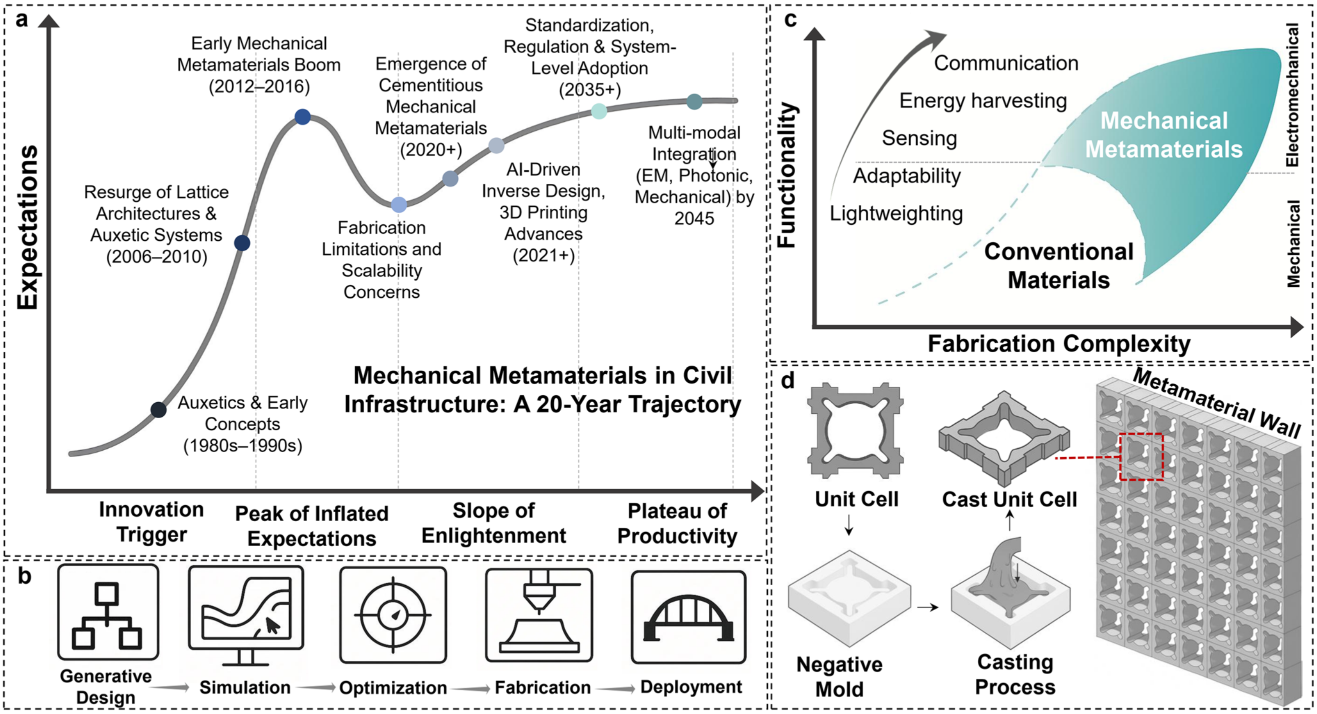


Figure 3. Opportunities and challenges in implementing mechanical metamaterials for civil infrastructure. a) Gartner Hype Cycle showing a 20-year projected trajectory for mechanical metamaterials in civil infrastructure. The timeline reflects a shift from early auxetic concepts to multifunctional and AI-driven applications, with emerging focus areas including cementitious systems, fabrication scale-up, regulatory integration, and multimodal metamaterial integration. b) Full digital workflow for infrastructure-focused metamaterial design and implementation, spanning generative design, simulation, optimization, fabrication, and deployment. c) Functionality versus fabrication complexity diagram comparing conventional materials and mechanical metamaterials. Added capabilities such as sensing, energy harvesting, and communication increase fabrication complexity, especially when multiple materials or integrated electronics are required. d) A casting-based fabrication method recently demonstrated by the author's team in refs. [7,8], showing reusable negative molds for producing cementitious metamaterial unit cells that can be assembled into modular walls. This approach enables high-fidelity replication of complex architectures while maintaining compatibility with standard construction workflows. Note: The modular wall shown in this figure was created through periodic assembly of a unit cell explored using the author's in-house generative metamaterial design algorithm.

their composition. This shift changes the design process entirely, as engineers move from selecting materials from standard charts to shaping internal structures tailored to specific performance goals. The main challenge is navigating a vast design space defined by unit cell geometries, feature sizes, and functional requirements.[133] Early design efforts relied heavily on intuition and physical prototyping; however, these trial-and-error methods quickly became impractical due to the combinatorial explosion of possibilities.[133] Topology optimization (TO) offered an early advancement by enabling property-driven material redistribution under defined constraints.[160] TO has been widely applied to the design of metamaterials.[161–169] At smaller scales in civil infrastructure components, TO can enable inverse design of target effective tensors and functions such as anisotropic stiffness, energy absorption, auxetic response, and bandgaps, while enforcing manufacturability constraints relevant to civil production (minimum feature size, printable overhangs, cover for reinforcement, demolding, and tolerances). Robust and reliability-based TO formulations can incorporate geometric and material variability to control stress ranges and fatigue demand. At the component scale, hierarchical TO can potentially map graded unit cell libraries onto panelization and joint layouts, improving constructability and inspection access in modular systems. However, TO remains limited by its reliance on simplified physics models, high computational cost, and its tendency to generate one solution at a time, making it inefficient for navigating large, multi-objective design spaces.[133,170,171]

Recent advances in generative models, including data-driven, physics-informed, and diffusion- or transformer-based AI architectures, offer promising avenues for efficiently sampling complex metamaterial design spaces.[126–138] The major concern about these approaches is that they rely on large training datasets and face difficulties when generalizing to unseen or novel geometries, particularly in the presence of nonlinear or multifunctional objectives.[172–174] One potential solution involves incorporating transfer learning and geometry-aware latent space representations that can generalize across different unit cell families and reduce dependence on exhaustive datasets. To further improve design efficiency and robustness, hybrid frameworks that integrate physics-based constraints with adaptive learning strategies are necessary. In contrast, evolutionary algorithms rooted in generative AI offer a more adaptive and exploratory alternative.[7,8] These algorithms can support multi-objective optimization, such as maximizing stiffness while minimizing weight, but they are computationally expensive. To address this, surrogate modeling techniques (e.g., neural networks or Gaussian processes) can be embedded within the evolutionary loop to approximate performance evaluations, significantly reducing the computational load.

Another key limitation is defining suitable objective functions for mechanical performance, especially in cases involving multilayered or multifunctional metamaterials. These systems often involve nonlinear behaviors and anisotropic responses that are difficult to encode using traditional metrics. To overcome this, hybrid objective formulations combining empirical data, finite element simulations, and physics-based constraints can guide optimization more effectively across hierarchical design spaces. For civil infrastructure, generative design offers a promising path toward faster modular construction. It allows for the creation of indexed libraries of unit cells that can be structurally tailored to specific components such as beams, decks, panels, and tunnel linings. This modular approach supports scalable integration into architectural forms and enables localized tuning based on site-specific stress conditions. Coupled with the continued rise of digital fabrication, these AI-enhanced workflows are increasingly compatible with end-to-end production pipelines from inverse design to automated large-scale manufacturing. As illustrated in Figure 3b, the process spans from inverse design through simulation, optimization, and fabrication, ultimately leading to real-world deployment.

6.2. Fabrication Precision and Scale

Despite success in laboratory settings,[4,24,145–147] implementing mechanical metamaterials in real infrastructure is still hindered by fabrication constraints. Many of the intricate geometries key to metamaterial performance are difficult to scale without sacrificing fidelity or cost-effectiveness. Additive manufacturing has enabled high-resolution production of polymeric and metallic architectures, but throughput remains limited for large-scale civil applications. Traditional materials such as concrete, steel, and composites present additional integration challenges. For instance, cementitious mechanical metamaterials require novel casting strategies (e.g., reusable molds, sacrificial cores, or hybrid casting-printing processes) to realize internal architectures at scale.[7,8] As illustrated in Figure 3c, advancing from basic mechanical functionalities such as lightweighting and adaptability to more complex capabilities like sensing, energy harvesting, and communication significantly increases fabrication complexity. For single base material, prefabrication of modular metamaterial units under controlled conditions followed by onsite assembly is a practical approach. These modules can be manufactured as tiles, blocks, or panels with embedded functionality. Figure 3d illustrates a casting-based workflow that employs reusable negative molds to fabricate concrete metamaterial unit cells, which are subsequently assembled into large-scale metamaterial walls.[8] This process enables repeatable, high-fidelity production of architected cementitious structures while remaining compatible with standard construction logistics.

Transitioning from purely mechanical to electromechanical capability is challenging, as it requires heterogeneous integration of multiple materials and components at meter scale. Energy harvesting typically needs co-fabrication of dielectric and conductive phases,[4] which introduces material compatibility issues and multimaterial printing demands. Sensing and communication add embedded conductive networks, electromechanical transduction layers, and, where used, electronics that conventional construction does not readily accommodate. Hybrid fabrication provides a practical path to scalability. Promising routes include 3D-printed stay-in-place formwork with cast-in electrodes, cold-spray metallization on prepared skins, stencil or direct ink printing of conductive inks with laminated dielectrics, robotic filament winding or tow placement of conductive fibers, ultrasonic or thermal welding of triboelectric polymer layers, or spray-applied overlays with embedded conductors. Success across these methods hinges on repeatable large-scale processing, reliable interfacial bonding, and electrical isolation from steel reinforcement to prevent stray currents and corrosion. Surface profiling, primers, or compliant interlayers help maintain adhesion and prevent delamination. Together, these techniques outline a realistic route to multifunctional, full-scale components capable of energy harvesting and sensing. Looking ahead, voxel-based robotic construction,[175] and standardized prefabrication of multifunctional metamaterial elements could offer scalable solutions that maintain strength, support multiple functions, and fit within current construction practices. However, initial fabrication costs for metamaterial components may exceed those of conventional materials. Modular production, reusable molds, and integrated functionality can offset these expenses through reduced mass, faster installation, and lower maintenance over the service life. Systematic pilot manufacturing and field trials should verify scalability, quantify variability and durability under service conditions, and provide cost and reliability data needed for adoption.

6.3. Integration with Traditional Infrastructure and Standards

The successful adoption of mechanical metamaterials depends not only on performance but on their compatibility with conventional civil infrastructure systems. Interfacing metamaterials with standard beams, columns, joints, or slabs introduces mechanical, regulatory, and practical challenges. Mechanical mismatch at joints or interfaces can lead to stress concentrations, unpredictable load transfer, or failure at connections. Novel joints, anchors, and couplers must be designed to respect the unique properties of mechanical metamaterials, especially their directional stiffness or compliant behavior. Retrofitting applications face similar issues when augmenting aging infrastructure with damping or sensing layers. Moreover, the regulatory environment is not yet prepared for these materials. Current design codes assume homogeneity and linear elastic behavior,[176–178] which many metamaterials do not satisfy. Their response is geometry dependent, often nonlinear and anisotropic, and may use local instability as a functional mechanism. In the absence of recognized standards, engineers lack tools to assess safety and building officials cannot approve designs. New guidelines, performance metrics, and test methods are needed through coordination with ASCE, ACI, and ASTM. A practical near-term path is a performance-based framework that explicitly treats anisotropy, nonlinearity, and variability. Early steps include defining representative volume elements and directional property cards, adapting existing tests to architecture-sensitive behavior, and establishing statistical allowables that capture manufacturing scatter.

These data should inform performance functions, limit states, and resistance factors suited to architected systems. This work can be shared among organizations, with ASTM focusing on method development, ASCE/SEI on design frameworks, and ACI on cementitious applications, leading to prestandards and pilot validation.

In parallel, uncertainty quantification must be a core element of qualification and certification.[179–182] Variability in base materials, geometry, and interfaces affects directional response and failure modes.[86] Directional property cards reporting scatter, minimum unit-cell counts and orientations, and interlaboratory comparisons can establish reliable design data. Validation protocols must also reflect geometry-sensitive behavior, since a material's strength may no longer be captured by a compressive cube test. Failure can localize at nodes, buckling zones, or within functional layers.[86,98] Durability under cyclic loading and environmental exposure is a another barrier. Quasi-static behavior defines the initial strength and stiffness of mechanical metamaterials, whereas time-dependent and environmental effects govern their service life. The same geometric features that enable function can elevate local stresses, hasten fatigue, or interact unpredictably with freeze–thaw, corrosion, and creep. Mitigation requires targeted aging and fatigue programs and digital twins with explicit geometry to bridge laboratory and field response. A design-for-test workflow should link unit-cell, panel, and system scales under combined mechanical and environmental actions. Design provisions should couple these protocols, define acceptable variability in systems based on metamaterials, and set durability targets tied to field-relevant loading histories.

6.3.1. Pathways for Deployment and Prioritization of Metamaterial Pilot

For near-term deployment, modular, compression-dominated components that require minimal or no steel reinforcement are the most suitable candidates. Compression-dominated modular metamaterial units can be prefabricated, proof-loaded, and bonded in the factory under controlled conditions, then replaced in service if needed. Pilot studies can focus on acoustic full-concrete noise barriers, delivered either as monolithic architected concrete panels or as factory-made sandwich panels with architected concrete or polymer cores and thin skins. Other suitable examples include segmental retaining wall blocks or mechanically stabilized earth facings, formed from interlocking precast blocks with internal cellular cores that provide confinement and crush control while requiring minimal reinforcement. Interior non-load-bearing wall and ceiling panels also represent promising applications, where architected cores can provide stiffness and acoustic control with simple fastening and negligible steel requirements. To complement these pilot opportunities, a structured framework is needed to guide prioritization, investment, and risk evaluation. While a complete cost-benefit-risk assessment requires field data, in the absence of extensive deployment histories, a first-order yet defensible prioritization can be achieved through the use of bounded parameter ranges, analogies to precast and modular construction, and explicit incorporation of uncertainty. A transparent scoring framework with documented weights and conservative priors can support this process, followed by sensitivity checks to evaluate robustness. Consistent with established public-sector benefit-cost and risk management frameworks (e.g., refs. [183–186]), following criteria can be defined for preliminary evaluation: (1) initial capital cost difference relative to a conventional alternative, (2) projected operation and maintenance cost savings during the first three years of service, (3) consequence of failure, (4) regulatory and code readiness for permitting, certification, and acceptance, (5) inspection and maintenance accessibility, (6) suitability for modular delivery, (7) environmental exposure severity, and (8) time to first pilot deployment. Criterion (1) quantifies the incremental upfront cost of procuring and installing the metamaterial-based alternative relative to a functionally equivalent conventional design, incorporating fabrication, transport, installation, and contingency. Criterion (7) captures the anticipated intensity of environmental degradation mechanisms over the service life, including moisture cycling, chloride and sulfate exposure, freeze–thaw effects, temperature fluctuations, ultraviolet radiation, abrasion, chemical exposure, and biofouling. Each criterion can be scored on a 1–5 scale (where 5 = favorable, 1 = difficult/risky) and combined using normalized weights. **Table 1** presents the prioritization matrix for near-term modular metamaterial pilots across civil infrastructure sectors. Since owner acceptance and regulatory approval timelines are strongly influenced by safety and certification factors, higher weighting should be assigned to consequence of failure and regulatory and code readiness.[185–187] As an illustrative baseline, the following weights can be used: consequence of failure (0.25), regulatory and code readiness (0.20), operation and maintenance cost savings (0.15), initial capital cost difference (0.15), inspection and maintenance access (0.10), suitability for modular delivery (0.10), environmental exposure severity (0.03), and time to first pilot deployment (0.02). These numbers yield approximate composite scores of 4.6, 3.4, 3.3, 2.1, and 1.1 for Architectural/Interior, Transportation, Buildings, Underground, and Offshore applications, respectively. The presented eight criteria serve as a practical initial framework. Additional criteria may be incorporated, and the weighting factors can be modified by agencies or asset owners to align with local priorities, regulatory context, and risk tolerance.

7. Conclusions

This perspective has presented mechanical metamaterials as a paradigm shift in how civil infrastructure is conceived, designed, and built. By overcoming the limitations of conventional materials, which are typically monolithic and compositionally fixed, these architected systems introduce pathways for lightweight structures, tunable mechanical properties, and integrated functionality. Rather than simply carrying loads, infrastructure components made from mechanical metamaterials can interact with their environment, respond to external stimuli, and adapt over time. The paper demonstrates how mechanical metamaterials can enhance the performance of components across all major infrastructure sectors. Applications range from reducing mass in bridge decks and tunnel linings, embedding energy harvesting into pavements, and enhancing acoustic management in building envelopes. These are not incremental upgrades but part of a broader shift toward systems that combine structural and

Table 1. Prioritization matrix for near-term modular metamaterial pilots across civil infrastructure sectors.

Sector	Representative modular pilot	Consequence of failure	Regulatory and code readiness	Operation and maintenance cost savings (1–3 years)	Initial capital cost difference	Inspection and maintenance access	Suitability for modular delivery	Environmental exposure severity	Time to pilot (Months)	Priority rank
Architectural interior	Interior non-load-bearing wall and ceiling panels	Low	Low	Moderate	Low to moderate	Easy	High	Low	6–12	1
Transportation	Modular acoustic noise-barrier panels	Medium	Medium	Moderate to high	Moderate	Moderate	High	Medium	12–24	2
Buildings	Panelized floor or wall retrofit modules for vibration and acoustics	Medium	Medium	Moderate	Moderate	Moderate	High	Low to medium	12–18	3
Underground	Segmental retaining wall blocks	High	Medium to high	Moderate	Moderate	Moderate to hard	Medium to high	Medium	18–24	4
Offshore	Splash-zone cladding or modular fairings for damping	Very high	High	Case dependent	High	Hard	Low to medium	Very high	>24	5

advanced functionalities. Within this framework, metamaterial-enabled civil infrastructure components are no longer passive elements. They can function as active systems that sense strain, monitor environmental conditions, and transmit information through their built-in mechanisms. Columns can signal fatigue, façades can react to thermal shifts, and pavements can convert mechanical loading into electrical signals. Each element contributes to a distributed system that interprets and responds to real-time conditions. The developments outlined in this perspective paper are supported by the convergence of three trends: geometry-driven material design, multifunctional integration, and scalable fabrication. Together, these advances redefine the role of materials in infrastructure, merging form and function into systems designed for active performance. To realize this vision, progress is needed in design automation, manufacturing, and implementation standards. Computational tools must efficiently explore complex design spaces. Fabrication must scale while preserving precision and cost-effectiveness. Continued research should prioritize durability metrics, geometry-sensitive validation methods, and fatigue characterization frameworks to support reliable long-term performance. Integration with existing construction workflows and codes remains essential. Cost realism requires systematic evaluation. Although initial fabrication costs may be higher, reduced mass, accelerated installation, and integrated multifunctionality can decrease life-cycle expenditures. Uncertainty quantification should be integrated into qualification and certification through directional property cards with stated scatter, representative unit-cell counts, proof tests at panel or module scale, and reliability factors based on measured variability. Pilot projects should use predefined performance and durability metrics, report uncertainty bounds, and publish cost data to create transferable allowables, procurement templates, and underwriting evidence. In addition to the technical barriers discussed, adoption also depends on liability and insurability, contractor readiness, and procurement. Priorities include establishing performance-based acceptance protocols that support underwriting, developing targeted training and certification, and conducting transparent pilot projects. Addressing these factors, alongside progress in design automation, scalable fabrication, and codes, will accelerate translation to practice. With these elements aligned, mechanical metamaterials can provide a foundation for civil infrastructure systems that are stronger, more efficient, and inherently intelligent.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

civil infrastructure, mechanical metamaterials, multifunctionality, multimodal integration, specific strength

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[1] H. Wu, *Advanced Civil Infrastructure Materials: Science, Mechanics and Applications*, Woodhead Publishing, Sawston, Cambridge, UK **2006**.
[2] A. Soliman, G. Hafeez, E. Erkmen, R. Ganesan, M. Ouf, A. Hammad, O. Moselhi, *Mater. Today: Proc.* **2022**, *60*, 365.
[3] M. T. Tiza, S. Imoni, E. O. Akande, O. Mogbo, V. H. Jiya, C. Onuzulike, *Recent Prog. Mater.* **2024**, *6*, 1.
[4] K. Barri, Q. Zhang, J. Kline, W. Lu, J. Luo, Z. Sun, A. H. Alavi, *Adv. Mater.* **2023**, *35*, 2211027.
[5] M. M. Rana, K. Hossain, *Int. J. Pavement Res. Technol.* **2023**, *16*, 264.
[6] R. Moini, *RILEM Tech. Lett.* **2023**, *8*, 125.
[7] W. Lu, J. Luo, W. Meng, Q. Zhang, A. H. Alavi, presented at the 60th Society Engineering Science Annual Conf. (SES 2023), Minneapolis, MN **2023**.
[8] A. H. Alavi, W. Lu, J. Luo, W. Meng, Developing lightweight and high-performance metamaterial concrete. Impactful Resilient Infrastructure Science and Engineering (IRISE) Research Consortium (Report No. IRISE-23-P22-01-01), University of Pittsburgh **2025**.
[9] S. M. Easa, W. Y. Yan, *Infrastructures* **2019**, *4*, 28.
[10] M. P. S. Redekar, *Int. Res. J. Eng. Technol.* **2017**, *4*, 2142.
[11] E. T. Agrawal, J. F. Unger, P. S. Koutsourelakis, *Data-Centric Eng.* **2024**, *5*, 20.
[12] K. Jamali, V. Kaushal, M. Najafi, *Infrastructures* **2021**, *6*, 108.
[13] J. U. Surjadi, X. Li, X. Xiong, N. X. Fang, Y. Lu, *Adv. Eng. Mater.* **2019**, *21*, 1800864.
[14] P. Jiao, J. Mueller, J. R. Raney, X. Zheng, A. H. Alavi, *Nat. Commun.* **2023**, *14*, 6004.
[15] R. Craster, S. Guenneau, M. Kadic, M. Wegener, *Rep. Prog. Phys.* **2023**, *86*, 094501.
[16] K. Bertoldi, V. Vitelli, J. Christensen, M. Van Hecke, *Nat. Rev. Mater.* **2017**, *2*, 1.
[17] T. A. Schaedler, W. B. Carter, *Annu. Rev. Mater. Res.* **2016**, *46*, 187.
[18] T. Frenzel, M. Kadic, M. Wegener, *Science* **2017**, *358*, 1072.
[19] S. Alkunte, M. Gupta, M. Rajeshirke, N. More, M. Cheepu, A. Gupta, I. Fidan, *Processes* **2024**, *12*, 2252.
[20] P. Zhang, P. Yu, R. Zhang, X. Chen, H. Tan, *Int. J. Mech. Sci.* **2023**, *259*, 108616.
[21] R. Nam, M. Jakubinek, H. Niknam, M. Rahmat, B. Ashrafi, H. E. Naguib, *Mater. Des.* **2023**, *228*, 111835.
[22] H. Zhang, X. Xie, *J. Zhejiang Univ., Sci., A* **2011**, *12*, 593.
[23] Y. Fujino, D. Siringoringo, *Struct. Eng. Int.* **2013**, *23*, 248.
[24] J. A. Rosewitz, H. A. Choshali, N. Rahbar, *Cem. Concr. Compos.* **2019**, *96*, 252.
[25] L. Stepinac, J. Galić, A. P. Vassilopoulos, *Compos. Struct.* **2024**, *343*, 118304.
[26] Q. Tan, H. Zhu, S. Yang, X. Yang, L. Ou, *Materials* **2024**, *17*, 2245.
[27] S. Yanagisawa, W. Burger, *Handbook of Precast Segmental Tunnel Lining Systems*, CRC Press, Boca Raton, FL **2024**, pp. 588–628.
[28] S. Bulson, *Buried Structures: Static and Dynamic Strength*, CRC Press, Boca Raton, FL **1984**.
[29] R. K. Goel, B. Singh, J. Zhao, *Underground Infrastructures: Planning, Design, and Construction*, Butterworth-Heinemann, Oxford, UK **2012**.
[30] B. C. Gerwick, Jr., *Construction of Marine and Offshore Structures*, CRC Press, Boca Raton, FL **2007**.
[31] O. Kapliński, *Sustainability* **2023**, *15*, 5967.
[32] R. Ioannidis, *Spatial Planning and Architectural Design for the Integration of Civil Infrastructure into Landscapes: Inferences from Renewable Energy Works and Dams*, National Technical University of Athens, Greece **2022**.
[33] D. Whitney, E. Crawley, O. de Weck, S. Eppinger, C. Magee, J. Moses, D. Wallace, *Engineering Systems Monograph*, MIT Engineering Systems Division, Cambridge, MA **2004**.
[34] A. J. MacDonald, C. R. Barnett, *New Zealand Eng.* **1964**, *19*, 291.
[35] W. A. Al-Kutti, A. B. M. S. Islam, Z. A. Kazmi, M. Sodangi, F. Anwar, M. Nasir, K. S. Alotaibi, *Earthq. Struct* **2022**, *23*, 493.
[36] EduPackCES, "Material and Process Selection Charts," Granta Design Ltd., University of Cambridge, **2009**.
[37] A. Neville, C. Aitcin, *Mater. Struct.* **1998**, *31*, 111.
[38] H. G. Russell, B. A. Graybeal, *Ultra-High Performance Concrete: A State-of-the-Art Report for the Bridge Community (No. FHWA-HRT-13-060)*, U.S. Department of Transportation, Federal Highway Administration, Office of Infrastructure Research and Development, McLean, VA **2013**.
[39] R. Ullah, Y. Qiang, J. Ahmad, N. I. Vatin, M. A. El-Shorbagy, *Materials* **2022**, *15*, 4131.
[40] T. Ponnampalam, S. Navaratnam, J. Thamboo, G. Zhang, *J. Perform. Constr. Facil.* **2023**, *37*, 04023051.
[41] S. N. Pilathottathil, A. Rauf, *Sustainability* **2024**, *16*, 6837.
[42] B. Salazar, P. Aghdasi, I. D. Williams, C. P. Ostertag, H. K. Taylor, *Mater. Des.* **2020**, *196*, 109184.
[43] C. Morris, L. Bekker, C. Spadaccini, M. Haberman, C. Seepersad, *Adv. Eng. Mater.* **2019**, *21*, 1900163.
[44] X. Zhang, H. Zhang, Y. Wang, X. Shi, *Buildings* **2022**, *12*, 2112.
[45] M. Meloni, Q. Zhang, J. Cai, D. S. H. Lee, *Sustainable Cities Soc.* **2023**, *97*, 104740.
[46] M. Norman, K. Arjomandi, in *Proc. 2nd World Congress Civil, Structural, Environmental Engineering (CSEE)*, Avestia Publishing, **2017**, *130*, pp. 2371.
[47] C. Y. Park, Y. A. Lee, J. Jang, M. W. Han, *Sensors* **2023**, *23*, 2150.
[48] W. R. Thornton, M. W. Trethewey, K. P. Maynard, J. F. Sadler, *Sound Vib.* **2006**, *40*, 10.
[49] E. Z. Berglund, J. G. Monroe, I. Ahmed, M. Noghabaei, J. Do, J. E. Pesantez, J. Levis, *J. Infrastruct. Syst.* **2020**, *26*, 03120001.
[50] Q. Zhang, A. H. Alavi, *Active Passive Smart Struct. Integr. Syst.* **2023**, *12483*, 245.
[51] Q. Zhang, K. Barri, S. R. Kari, Z. L. Wang, A. H. Alavi, *Adv. Funct. Mater.* **2021**, *31*, 2105825.
[52] X. Xu, Q. Wu, Y. Pang, Y. Cao, Y. Fang, G. Huang, C. Cao, *Adv. Funct. Mater.* **2022**, *32*, 2107896.
[53] K. Barri, P. Jiao, Q. Zhang, J. Chen, Z. L. Wang, A. H. Alavi, *Nano Energy* **2021**, *86*, 106074.
[54] Y. Pang, X. Zhu, Y. Yu, S. Liu, Y. Chen, Y. Feng, *Nano Res.* **2022**, *15*, 5450.
[55] G. Hu, C. Zhao, Y. Yang, X. Li, J. Liang, *Appl. Energy* **2022**, *306*, 118037.
[56] X. Zhou, K. Parida, J. Chen, J. Xiong, Z. Zhou, F. Jiang, S. Lee, *Adv. Energy Mater.* **2023**, *13*, 2301159.
[57] T. Fey, F. Eichhorn, G. Han, K. Ebert, M. Wegener, A. Roosen, P. Greil, *Smart Mater. Struct.* **2015**, *25*, 015017.
[58] Y. Li, E. Baker, T. Reissman, C. Sun, W. K. Liu, *Appl. Phys. Lett.* **2017**, *111*, 251903.
[59] H. J. Lee, D. - K. Kang, J. Kang, C. Lee, M. D. Bartlett, J. Yeo, H. E. Jeong, *Adv. Funct. Mater.* **2025**, *35*, 2506293.
[60] J. Luo, W. Lu, P. Jiao, D. Jang, K. Barri, J. Wang, Z. L. Wang, A. H. Alavi, *Mater. Today* **2025**, *83*, 145.

[61] H. Wang, J. Wang, K. Yao, J. Fu, X. Xia, R. Zhang, Y. Zi, *Sci. Adv.* **2021**, *7*, abi6751.
[62] H. Zhao, M. Xu, M. Shu, J. An, W. Ding, X. Liu, Z. L. Wang, *Nat. Commun.* **2022**, *13*, 3325.
[63] W. Zhu, W. Deng, Y. Liu, J. Lu, H. X. Wang, Z. K. Lin, Z. Liu, *Rep. Prog. Phys.* **2023**, *86*, 106501.
[64] Y. Jin, L. He, Z. Wen, B. Mortazavi, H. Guo, D. Torrent, Y. Li, *Nanophotonics* **2022**, *11*, 439.
[65] G. Ma, P. Sheng, *Sci. Adv.* **2016**, *2*, 1501595.
[66] S. Dogra, A. Gupta, *Acoustics* **2021**, *3*, 630.
[67] C. A. H. Owens, *Doctoral Dissertation*, University of California, Irvine, Irvine, CA **2025**.
[68] J. Qi, Z. Chen, P. Jiang, W. Hu, Y. Wang, Z. Zhao, D. Fang, *Adv. Sci.* **2022**, *9*, 2102662.
[69] Q. Sun, G. Zhi, S. Zhou, R. Tao, J. Qi, *Chin. J. Mech. Eng.* **2024**, *37*, 123.
[70] *Metamaterials: Physics and Engineering Explorations* (Eds: N. Engheta, R. W. Ziolkowski), Wiley, Hoboken, NJ **2006**.
[71] T. J. Cui, D. R. Smith, R. Liu, *Metamaterials*, Springer,New York **2010**, p. 1.
[72] I. F. Akyildiz, Z. Sun, M. C. Vuran, *Phys. Commun.* **2009**, *2*, 167.
[73] S. S. Shete, R. A. Vatti, K. V. Raigandhi, in *2017 Fourth Int. Conf. Image Information Processing (ICIIP)*, IEEE Shimla, India **2017**, pp. 1–4.
[74] S. E. Lapinsky, A. C. Easty, *J. Crit. Care* **2006**, *21*, 267.
[75] X. E. Zhao, *App. Mech. Mater.* **2014**, *596*, 918.
[76] A. K. Sisodia, *Int. J. Sci. Res. Arch.* **2025**, *14*, 954.
[77] P. Singh, C. Mutzel, S. MacNaughton, S. Sonkusale, *Prog. Electromagn. Res.* **2013**, *141*, 117.
[78] W. Cai, V. M. Shalaev, *Optical Metamaterials: Fundamentals and Applications*, Springer, Berlin **2010**.
[79] D. R. Smith, J. B. Pendry, M. C. Wiltshire, *Science* **2004**, *305*, 788.
[80] D. E. Parker, *Wiley Interdiscip. Rev.: Climate Change* **2010**, *1*, 123.
[81] R. S. Lakes, *Science* **1987**, *235*, 1038.
[82] K. E. Evans, M. A. Nkansah, I. J. Hutchinson, S. C. Rogers, *Nature* **1991**, *353*, 124.
[83] G. W. Milton, *J. Mech. Phys. Solids* **1992**, *40*, 1105.
[84] R. Lakes, *Nature* **1993**, *361*, 511.
[85] A. G. Evans, J. W. Hutchinson, M. F. Ashby, *Prog. Mater. Sci.* **1999**, *43*, 171.
[86] L. J. Gibson, M. F. Ashby, *Cellular Solids: Structure and Properties*, Cambridge University Press,Cambridge, UK **1997**.
[87] V. S. Deshpande, M. F. Ashby, N. A. Fleck, *Acta Mater.* **2001**, *49*, 1035.
[88] H. B. G. Wadley, N. A. Fleck, A. G. Evans, *Compos. Sci. Technol.* **2003**, *63*, 2331.
[89] L. Valdevit, A. Pantano, H. A. Stone, A. G. Evans, *Int. J. Heat Mass Transf.* **2006**, *49*, 3819.
[90] A. J. Jacobsen, W. B. Carter, S. Nutt, *Adv. Mater.* **2007**, *19*, 3892.
[91] O. Cansizoglu, O. Harrysson, D. Cormier, H. West, T. Mahale, *Mater. Sci. Eng., A* **2008**, *492*, 468.
[92] H. N. G. Wadley, *Phil. Trans. R. Soc. A* **2006**, *364*, 31.
[93] K. Bertoldi, M. Reis, S. Willshaw, T. Mullin, *Adv. Mater.* **2010**, *22*, 361.
[94] G. N. Greaves, A. L. Greer, R. S. Lakes, T. Rouxel, *Nat. Mater.* **2011**, *10*, 823.
[95] T. A. Schaedler, A. J. Jacobsen, A. Torrents, A. E. Sorensen, J. Lian, J. R. Greer, L. Valdevit, W. B. Carter, *Science* **2011**, *334*, 962.
[96] L. Valdevit, A. J. Jacobsen, J. R. Greer, W. B. Carter, *J. Am. Ceram. Soc.* **2011**, *94*, S15.
[97] M. Kadic, T. Bückmann, R. Schittny, M. Wegener, *Appl. Phys. Lett.* **2012**, *100*, 191901.
[98] X. Zheng, H. Lee, T. H. Weisgraber, M. Shusteff, J. DeOtte, E. B. Duoss, J. D. Kuntz, M. M. Biener, Q. Ge, J. A. Jackson, S. O. Kucheyev, N. X. Fang, C. M. Spadaccini, *Science* **2014**, *344*, 1373.
[99] L. R. Meza, S. Das, J. R. Greer, *Science* **2014**, *345*, 1322.
[100] J. Christensen, M. Kadic, M. Wegener, O. Kraft, *MRS Commun.* **2015**, *5*, 453.
[101] A. Rafsanjani, A. Akbarzadeh, D. Pasini, *Adv. Mater.* **2015**, *27*, 5931.
[102] L. R. Meza, A. J. Zelhofer, N. Clarke, A. J. Mateos, D. M. Kochmann, J. R. Greer, *Proc. Natl. Acad. Sci. USA* **2015**, *112*, 11502.
[103] X. Li, H. Gao, *Nat. Mater.* **2016**, *15*, 373.
[104] A. A. Zadpoor, *Mater. Horiz.* **2016**, *3*, 371.
[105] X. Zheng, W. Smith, J. Jackson, B. Moran, H. Cui, D. Chen, J. Ye, N. Fang, N. Rodriguez, T. Weisgraber, C. M. Spadaccini, *Nat. Mater.* **2016**, *15*, 1100.
[106] B. Haghpanah, L. Salari-Sharif, P. Pourrajab, J. Hopkins, L. Valdevit, *Adv. Mater.* **2016**, *28*, 7915.
[107] J. T. B. Overvelde, T. A. de Jong, Y. Shevchenko, S. A. Becerra, G. M. Whitesides, J. C. Weaver, C. Hoberman, K. Bertoldi, *Nat. Commun.* **2016**, *7*, 10929.
[108] J. Bauer, L. R. Meza, T. A. Schaedler, R. Schwaiger, X. Zheng, L. Valdevit, *Adv. Mater.* **2017**, *29*, 1701850.
[109] H. M. A. Kolken, A. A. Zadpoor, *RSC Adv.* **2017**, *7*, 5111.
[110] S. Kumar, S. Tan, L. Zheng, D. M. Kochmann, *npj Comput. Mater.* **2020**, *6*, 73.
[111] P. Jiao, H. Hasni, N. Lajnef, A. H. Alavi, *Mater. Des.* **2020**, *187*, 108214.
[112] H. Tao, J. Gilbert, *Adv. Funct. Mater.* **2020**, *30*, 2001720.
[113] Z. Ren, L. Ji, R. Tao, M. Chen, Z. Wan, Z. Zhao, D. Fang, *Extreme Mech. Lett.* **2021**, *42*, 101077.
[114] C. E. Helou, R. Buskohl, C. E. Tabor, R. L. Harne, *Nat. Commun.* **2021**, *12*, 1633.
[115] T. Mei, Z. Meng, K. Zhao, C. Chen, *Nat. Commun.* **2021**, *12*, 7234.
[116] F. Zangeneh-Nejad, D. L. Sounas, A. Alu, R. Fleury, *Nat. Rev. Mater.* **2021**, *6*, 207.
[117] K. Barri, Q. Zhang, I. Swink, Y. Aucie, K. Holmberg, R. Sauber, A. H. Alavi, *Adv. Funct. Mater.* **2022**, *32*, 2203533.
[118] H. Cui, D. Yao, R. Hensleigh, H. Lu, A. Calderon, Z. Xu, S. Davaria, Z. Wang, X. Zheng, *Science* **2022**, *376*, 1287.
[119] R. H. Lee, E. A. B. Mulder, J. B. Hopkins, *Sci. Rob.* **2022**, *7*, abq7278.
[120] Q. Zhang, K. Barri, Jiao, W. Lu, J. Luo, W. Meng, J. Wang, L. Hong, J. Mueller, Z. L. Wang, A. H. Alavi, *Mater. Today* **2023**, *65*, 78.
[121] J. Chen, X. Miao, H. Ma, J. B. Hopkins, G. Huang, *Mater. Des.* **2024**, *244*, 113093.
[122] H. Chen, J. Shi, L. Yan, N. Keena, A. Akbarzadeh, *Nano Energy* **2025**, *140*, 111001.
[123] H. Chen, J. Shi, L. Yan, A. Akbarzadeh, *Adv. Funct. Mater.* **2024**, *35*, 2416749.
[124] J. Shi, K. Ju, H. Chen, V. Orsat, A. P. Sasmito, A. Ahmadi, A. Akbarzadeh, *Adv. Funct. Mater.* **2024**, *35*, 2417618.
[125] J. B. Hopkins, *J. Intell. Mater. Syst. Struct.* **2025**, *36*, 1266.
[126] H. T. Kollmann, D. W. Abueidda, S. Koric, E. Guleryuz, N. A. Sobh, *Mater. Des.* **2020**, *196*, 109098.
[127] C. Chen, G. Gu, *Adv. Sci.* **2020**, *7*, 1902607.
[128] P. Jiao, A. H. Alavi, *Int. Mater. Rev.* **2021**, *66*, 365.
[129] Z. Xu, Z. Fan, W. Pang, Y. Zi, Y. Zhang, *Extreme Mech. Lett.* **2021**, *51*, 101582.
[130] B. Deng, A. Zareei, X. Ding, J. C. Weaver, C. H. Rycroft, K. Bertoldi, *Adv. Mater.* **2022**, *34*, 2206238.
[131] A. E. Forte, Z. Hanakata, L. Jin, E. Zari, A. Zareei, M. C. Fernandes, L. Sumner, J. Alvarez, K. Bertoldi, *Adv. Funct. Mater.* **2022**, *32*, 2111610.
[132] C. S. Ha, D. Yao, Z. Xu, C. Liu, H. Liu, D. Elkins, M. Kile, V. Deshpande, Z. Kong, X. Zheng, *Nat. Commun.* **2023**, *14*, 5765.
[133] Q. Zhang, H. Yu, K. Barri, Z. Wan, W. Lu, J. Luo, A. H. Alavi, *Adv. Intell. Syst.* **2023**, *5*, 2300019.
[134] C. Mo, P. Perdikaris, J. R. Raney, *J. Eng. Mech.* **2023**, *149*, 04023032.
[135] M. Maurizi, D. Xu, Y. T. Wang, D. Yao, D. Hahn, M. Oudich, A. Satpati, Z. Wang, X. R. Zheng, *Nat. Mach. Intell.* **2025**, *7*, 1023.

[136] D. Lee, W. Chen, W. (Wayne) Chen, L. Wang, Y. Chan, W. Chen, *Int. J. Numer. Methods Eng.* **2023**, *36*, 2305254.
[137] W. Chen, W. W. Chen, R. Sun, D. Lee, C. M. Portela, W. Chen, *Adv. Mater.* **2024**, *7*, 2400611.
[138] P. Jiao, C. Zhang, W. Meng, J. Wang, D. Jang, Z. Wu, N. Agarwal, A. H. Alavi, *ACS Appl. Mater. Interfaces* **2025**, *17*, 2991.
[139] S. Brûlé, E. H. Javelaud, S. Enoch, S. Guenneau, *Phys. Rev. Lett.* **2014**, *112*, 133901.
[140] S. Krödel, N. Thomé, C. Daraio, *Extreme Mech. Lett.* **2015**, *4*, 111.
[141] M. Miniaci, A. Krushynska, F. Bosia, N. M. Pugno, *New J. Phys.* **2016**, *18*, 083041.
[142] A. Colombi, S. Guenneau, P. Roux, R. V. Craster, *Sci. Rep.* **2016**, *6*, 25320.
[143] A. Palermo, S. Krödel, A. Marzani, C. Daraio, *Sci. Rep.* **2016**, *6*, 1.
[144] O. Casablanca, G. Ventura, F. Garesci, B. Azzerboni, B. Chiaia, M. Chiappini, G. Finocchio, *J. Appl. Phys.* **2018**, *123*, 174903.
[145] T. Zahra, M. Dhanasekar, *Constr. Build. Mater.* **2017**, *147*, 143.
[146] M. Moini, J. Olek, J. P. Youngblood, B. Magee, D. Zavattieri, *Adv. Mater.* **2018**, *30*, 1802123.
[147] P. Aghdasi, I. D. Williams, B. Salazar, N. Panditi, H. K. Taylor, C. P. Ostertag, *Compos. Struct.* **2019**, *207*, 373.
[148] S. M. Sajadi, J. Boul, C. Thaemlitz, A. K. Meiyazhagan, A. B. Puthirath, C. S. Tiwary, M. Ajayan, *Adv. Eng. Mater.* **2019**, *21*, 1801380.
[149] V. Nguyen-Van, P. Tran, C. Peng, L. Pham, G. Zhang, H. Nguyen-Xuan, *Autom. Constr.* **2020**, *119*, 103324.
[150] Y. Xu, B. Šavija, *Heron* **2021**, *66*, 239.
[151] G. Tzortzinis, A. Gross, S. Gerasimidis, *Extreme Mech. Lett.* **2022**, *52*, 101681.
[152] R. Xia, X. Song, L. Sun, W. Wu, C. Li, T. Cheng, G. Qian, *Phys. Status Solidi B* **2018**, *255*, 1700343.
[153] P. Jiao, A. H. Alavi, *Mater. Des.* **2019**, *162*, 405.
[154] K. Meena, S. Singamneni, *Mater. Des.* **2019**, *173*, 107779.
[155] P. Jiao, A. H. Alavi, *Appl. Mater. Today* **2020**, *21*, 100816.
[156] C. D. Pierce, C. L. Willey, V. W. Chen, J. O. Hardin, J. D. Berrigan, A. T. Juhl, K. H. Matlack, *Smart Mater. Struct.* **2020**, *29*, 065004.
[157] Y. Zhang, Q. Wang, M. Tichem, F. vab Keulen, *Extreme Mech. Lett.* **2020**, *34*, 100593.
[158] H. Fan, L. Yang, Y. Tian, Z. Wang, *Compos. Struct.* **2020**, *243*, 112244.
[159] T. Vitalis, A. Gross, G. Tzortzinis, B. Schagen, S. Gerasimidis, *Constr. Build. Mater.* **2024**, *457*, 139165.
[160] Y. Liu, Y. Wang, H. Ren, Z. Meng, X. Chen, Z. Li, L. Wang, W. Chen, Y. Wang, J. Du, *Nat. Commun.* **2024**, *15*, 2984 .
[161] A. Nale, A. Chiozzi, *Int. J. Mech. Sci.* **2025**, *295*, 110204.
[162] A. Alhembar, A. N. Alagha, M. M. Naji, J. Sheikh-Ahmad, F. Jarrar, *Adv. Eng. Mater.* **2023**, *25*, 2201780.
[163] Y. Lu, Q. Luo, L. Tong, *Comput. Methods Appl. Mech. Eng.* **2025**, *434*, 117587.
[164] Y. Jia, K. Liu, X. S. Zhang, *Comput. Methods Appl. Mech. Eng.* **2024**, *425*, 116864.
[165] N. Shi, W. Zhang, H. Liu, F. Meng, L. Zhao, *Mater. Today* **2024**, *425*, 13.
[166] C. Zhao, C. X. Liu, G. L. Yu, *J. Comput. Des. Eng.* **2024**, *11*, 56.
[167] P. Zhou, S. Wan, X. Wang, A. Song, R. Yue, *Comput. Geotech.* **2023**, *164*, 105774.
[168] Z. Liu, H. W. Dong, G. L. Yu, L. Cheng, *Compos. Struct.* **2022**, *295*, 115863.
[169] P. Dong, J. Hu, C. Lin, W. Ding, J. Liu, Y. Liu, *Mater. Des.* **2024**, *244*, 113155.
[170] Y. M. Xie, *Archit. Intell.* **2022**, *1*, 2.
[171] C. Xiang, A. Chen, H. Li, D. Wang, B. Ge, H. Chang, *Sci. Rep.* **2025**, *15*, 9350.
[172] L. Wang, Y. C. Chan, F. Ahmed, Z. Liu, Zhu, W. Chen, *Comput. Methods Appl. Mech. Eng.* **2020**, *372*, 113377.
[173] X. Zheng, X. Zhang, T. T. Chen, I. Watanabe, *Adv. Mater.* **2023**, *35*, 2302530.
[174] H. Park, Z. Li, A. Walsh, *Matter* **2024**, *7*, 2355.
[175] M. S. Khan, H. S. Cho, J. Seo, *Appl. Sci.* **2025**, *15*, 7899.
[176] American Concrete Institute, *Building Code Requirements for Structural Concrete (ACI 318-19) and Commentary*, American Concrete Institute,Farmington Hills, MI **2019**.
[177] J. G. MacGregor, J. K. Wight, *Reinforced Concrete: Mechanics and Design*, 6th ed., Pearson Education, Upper Saddle River, NJ **2012**.
[178] A. H. Nilson, D. Darwin, C. W. Dolan, *Design of Concrete Structures*, 14th ed., McGraw-Hill Education, New York, NY **2010**.
[179] C.-J. Bae, A. B. Diggs, A. Ramachandran, *Addit. Manuf.* **2018**, 181.
[180] B.-H. Jones, R. E., B. L. Boyce, A. L. Frankel, N. Heckman, M. Khalil, J. T. Ostien, J. A. Templeton, *Uncertainty Quantification of Microstructural Material Variability Effects (No. SAND2019-11089)*, Sandia National Laboratories (SNL-NM), Albuquerque, NM **2019**.
[181] T. Nguyen, A. Ghazlan, T. Nguyen, H. T. Thai, T. Ngo, *Int. J. Mech. Sci.* **2020**, *173*, 105468.
[182] S. Mahadevan, P. Nath, Z. Hu, *ASCE-ASME J. Risk Uncertainty Eng. Syst, Part B: Mech. Eng.* **2022**, *8*, 010801.
[183] USDOT, *Benefit–Cost Analysis Guidance for Discretionary Grant Programs*, U.S. Department of Transportation, Washington, DC **2024**.
[184] OMB, *Circular A-94: Guidelines and Discount Rates for Benefit–Cost Analysis of Federal Programs*, Office of Management and Budget, Washington, DC **2023**.
[185] NCHRP Report 806. Guide to Cross-Asset Resource Allocation and the Impact on Transportation System Performance, Transportation Research Board, **2015**.
[186] FHWA, *Incorporating Risk Management into Transportation Asset Management Plans*, Federal Highway Administration, Washington, DC **2017**.
[187] ISO 31000:2018, Risk management, Guidelines. International Organization for Standardization.